Name

Mathematics Stage 5 Path

Coordinate Geometry Problems

Book 1 Solve problems by applying coordinate geometry formulas

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Syllabus Content

MA5-LIN-P-01 describes and applies transformations, the midpoint, gradient/slope and distance formulas, and equations of lines to solve problems

Solve problems by applying coordinate geometry formulas

- Solve problems including those involving geometrical figures by applying coordinate geometry formulas

Coordinate Geometry Problems: Triangles

ⓘ Types of Triangles and their Properties

Type of Triangle	Definition
Equilateral Triangle	Three sides of equal length.
Isosceles Triangle	Two sides of equal length.
Right-angled Triangle	Side lengths satisfy Pythagoras' Theorem **or** Gradients of two sides are negative reciprocals.
Scalene Triangle	All side lengths different.

ⓘ Coordinate Geometry Formulas

Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Parallel if $m_1 = m_2$

Perpendicular if $m_1 = -\frac{1}{m_2}$

Area of a Triangle

$$A = \frac{1}{2}bh$$

Where b and h are perpendicular to each other.

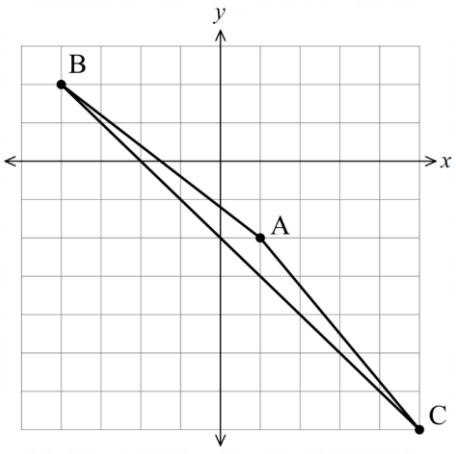
Example Solve coordinate geometry problems involving triangles

A triangle has vertices at $A(1, -2), B(-4, 2), C(5, -7)$.
Find the lengths of the sides AB and AC and hence show it is isosceles.

$$AB = \sqrt{(-4 - 1)^2 + \big(2 - (-2)\big)^2}$$
$$= \sqrt{41}$$

$$AC = \sqrt{(5 - 1)^2 + \big(-7 - (-2)\big)^2}$$
$$= \sqrt{41}$$

$AB = AC$, so ΔABC is isosceles.

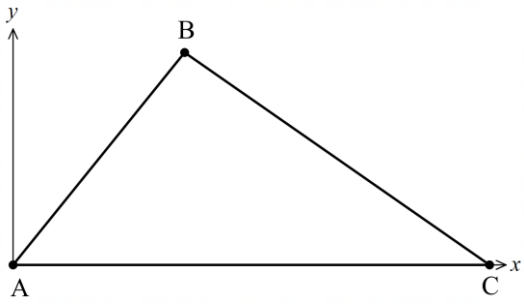


Guided Practice

The vertices of the ΔABC are given by $A(0,0), B(9,12)$ and $C(25,0)$.

a Find the lengths of the three sides AB, BC, AC.

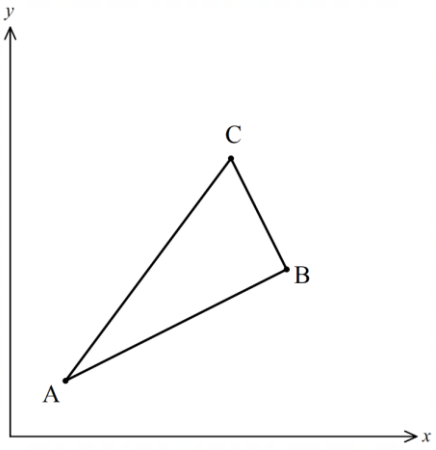
b By checking that $AB^2 + BC^2 = AC^2$, show that ΔABC is right-angled.



Triangle ABC has vertices at $A(1,1), B(5,3)$ and $C(4,5)$.

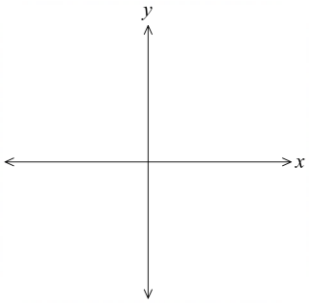
a Find the gradients of line segments AB and BC.

b Hence show the area of the triangle is approximately 3.35 u².



1 The triangle ABC has vertices A(−1, 0), B(3, 2) and C(4, 0).

Calculate the gradient of each side, and hence show that ΔABC is a right-angled triangle.



$m_{AB} = \frac{2-0}{3-(-1)} = \frac{1}{2}$

$m_{BC} = \frac{0-2}{4-3} = -2$

$m_{CA} = \frac{0-0}{4-(-1)} = 0$

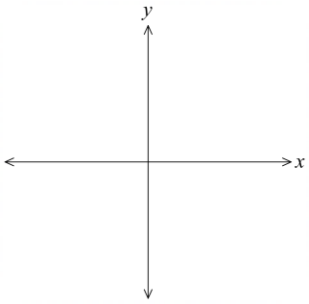
$m_{AB} \times m_{BC} = \frac{1}{2} \times (-2) = -1$

AB ⊥ BC, so Δ ABC is right-angled at B.

2

a Find the length of each side of the triangle formed by $P(0,3), Q(1,7)$ and $R(5,8)$.

b Hence show that ΔPQR is isosceles.



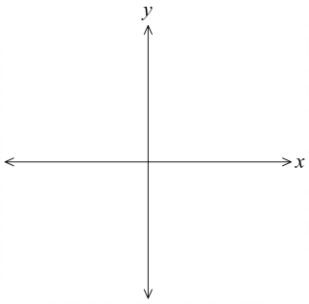
$PQ = \sqrt{(1-0)^2 + (7-3)^2} = \sqrt{17}$

$QR = \sqrt{(5-1)^2 + (8-7)^2} = \sqrt{17}$

$PR = \sqrt{(5-0)^2 + (8-3)^2} = \sqrt{50}$

$PQ = QR = \sqrt{17}$, so Δ PQR is isosceles.

3 Prove that $A(1,-1), B(-1,1), C\left(-\sqrt{3}, -\sqrt{3}\right)$ form an equilateral triangle.



$AB = \sqrt{(-1-1)^2 + \left(1-(-1)\right)^2} = \sqrt{8}$

$BC = \sqrt{\left(-\sqrt{3}-(-1)\right)^2 + \left(-\sqrt{3}-1\right)^2} = \sqrt{8}$

$CA = \sqrt{\left(1-(-\sqrt{3})\right)^2 + \left(-1-(-\sqrt{3})\right)^2} = \sqrt{8}$

AB = BC = CA = $\sqrt{8}$, so equilateral.

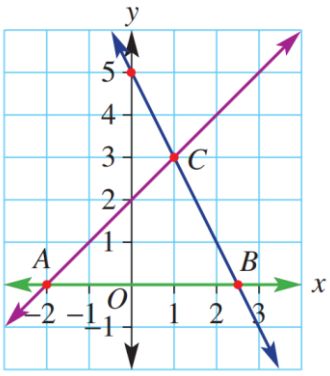
4 The three lines with equations $y = 0,\ y = x + 2$ and $y = -2x + 5$ are illustrated here.

a State the coordinates of the intersection point of $y = x + 2$ and $y = -2x + 5$.

Intersection point at (1,3)

b Use the formula for the area of a triangle to calculate the area of the enclosed region ΔACB

Area = $\frac{1}{2}bh = \frac{1}{2}(4.5)(3) = 6.75$ square units

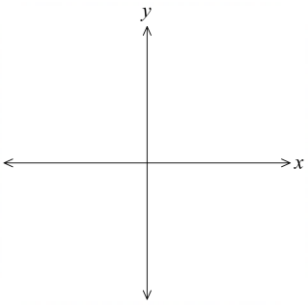


5 Consider a triangle ΔXYZ with vertices $X(0,-4), Y(4,2)$ and $Z(-2,6)$.

a Find the side lengths.

b Show that ΔXYZ is a right-angled isosceles triangle by showing that its side lengths satisfy Pythagoras' theorem.

c Hence find the area of ΔXYZ.



$XY = \sqrt{(4-0)^2 + (2-(-4))^2} = \sqrt{52}$

$YZ = \sqrt{(-2-4)^2 + (6-2)^2} = \sqrt{52}$

$XZ = \sqrt{(-2-0)^2 + (6-(-4))^2} = \sqrt{104}$

$XY^2 + YZ^2 = 52 + 52 = 104 = XZ^2$

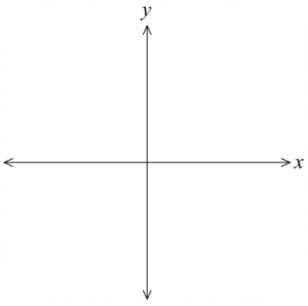
By Pythagoras' theorem, ΔXYZ is right-angled.

Area $= \frac{1}{2(XY)(YZ)} = \frac{1}{2}(2\sqrt{13})(2\sqrt{13}) = 26$ square units

6 The triangle ABC has vertices $A(-1,0), B(3,2)$ and $C(4,0)$.

a Calculate the gradient of each side,

b Explain why ΔABC is a right-angled triangle.



$m_{AB} = \frac{2-0}{3-(-1)} = \frac{1}{2}$

$m_{BC} = \frac{0-2}{4-3} = -2$

$m_{CA} = \frac{0-0}{-1-4} = 0$

$m_{AB} \times m_{BC} = \frac{1}{2} \times (-2) = -1$

AB ⊥ BC, so Δ ABC is right-angled at B.

7 Show that the triangle with vertices A(1, 0), B(6, 5) and C(0, 2) is right-angled by:

a Using Pythagoras' Theorem.

$AB^2 = (6-1)^2 + (5-0)^2 = 50$
$BC^2 = (0-6)^2 + (2-5)^2 = 45$
$CA^2 = (1-0)^2 + (0-2)^2 = 5$
$AB^2 = BC^2 + CA^2$
$50 = 45 + 5$
By Pythagoras' theorem, triangle is right-angled at C.

b Using the gradients of two sides.

$m_{BC} = \frac{2-5}{0-6} = \frac{1}{2}$
$m_{CA} = \frac{0-2}{1-0} = -2$
$m_{BC} \times m_{CA} = \frac{1}{2} \times (-2) = -1$
BC ⊥ CA, so triangle is right-angled at C.

8 The vertices of ΔABC are $A(0,0), B(3,4)$ and $C\left(\frac{25}{3}, 0\right)$

a Find the gradient of:

i AB

$$m_{AB} = \frac{4-0}{3-0} = \frac{4}{3}$$

ii BC

$$m_{BC} = \frac{0-4}{\frac{25}{3}-3} = -\frac{3}{4}$$

iii CA

$$m_{CA} = \frac{0-0}{0-\frac{25}{3}} = 0$$

b What type of triangle is ΔABC? Why?

$$m_{AB} \times m_{BC} = \frac{4}{3} \times \left(-\frac{3}{4}\right) = -1$$

AB ⊥ BC, so Δ ABC is right-angled at B.

c Find the perimeter of ΔABC.

$$AB = \sqrt{3^2 + 4^2} = 5$$

$$BC = \sqrt{\left(\frac{16}{3}\right)^2 + 4^2} = \frac{20}{3}$$

$$CA = \frac{25}{3}$$

$$\text{Perimeter} = 5 + \frac{20}{3} + \frac{25}{3} = 20 \text{ units}$$

9 Each set of three points given below forms a triangle that is either isosceles, equilateral, right-angled, or none of the above.
Find the side lengths of each triangle and hence determine its type.

a $A(-1,0),\ B(1,0),\ C\left(0,\sqrt{3}\right)$

$AB =$

$BC =$

$AC =$

$$AB = \sqrt{(1-(-1))^2 + (0-0)^2} = 2$$

$$BC = \sqrt{(0-1)^2 + \left(\sqrt{3}-0\right)^2} = 2$$

$$AC = \sqrt{(0-(-1))^2 + \left(\sqrt{3}-0\right)^2} = 2$$

Equilateral triangle (all sides equal)

b $P(-1,1),\ Q(0,-1),\ R(3,3)$

$$PQ = \sqrt{(0-(-1))^2 + (-1-1)^2} = \sqrt{5}$$

$$QR = \sqrt{((3-0)^2 + (3-(-1))^2)} = 5$$

$$PR = \sqrt{((3-(-1))^2 + (3-1)^2)} = \sqrt{20} = 2\sqrt{5}$$

$$PQ^2 + PR^2 = 5 + 20 = 25 = QR^2$$

Right − angled triangle at P

c $D(1,1),\ E(2,-2),\ F(-3,0)$

$$DE = \sqrt{(2-1)^2 + (-2-1)^2} = \sqrt{10}$$

$$EF = \sqrt{(-3-2)^2 + (0-(-2))^2} = \sqrt{29}$$

$$DF = \sqrt{(-3-1)^2 + (0-1)^2} = \sqrt{17}$$

scalene

d $X(-3,-1),\ Y(0,0),\ Z(-2,2)$

$$XY = \sqrt{(0-(-3))^2 + (0-(-1))^2} = \sqrt{10}$$

$$YZ = \sqrt{(-2-0)^2 + (2-0)^2} = \sqrt{8}$$

$$XZ = \sqrt{(-2-(-3))^2 + (2-(-1))^2} = \sqrt{10}$$

Isosceles

10 Find the gradients of the three lines $5x - 7y + 5 = 0,\ 2x - 5y + 7 = 0$ and $7x + 5y + 2 = 0$. Show that they enclose a right-angled triangle.

Line 1: $5x - 7y + 5 = 0 \rightarrow m_1 = \frac{5}{7}$

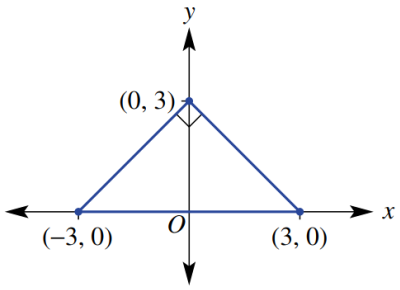
Line 2: $2x - 5y + 7 = 0 \rightarrow m_2 = \frac{2}{5}$

Line 3: $7x + 5y + 2 = 0 \rightarrow m_3 = -\frac{7}{5}$

$m_1 \times m_3 = \frac{5}{7} \times \left(-\frac{7}{5}\right) = -1$

Lines 1 and 3 are perpendicular, so triangle is right-angled.

11 A right-angled isosceles triangle has vertices at $A(0,3)$, $B(3,0)$ and $C(-3,0)$. Find the equation of each side.



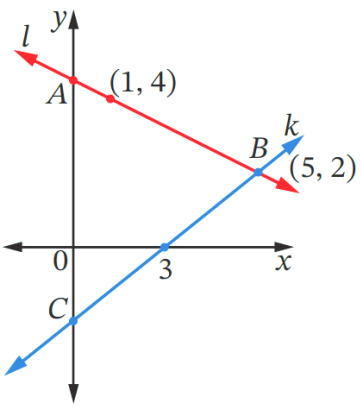
AB: $m = \frac{0-3}{3-0} = -1$, passes through $(0,3)$: $y = -x + 3$

AC: $m = \frac{0-3}{-3-0} = 1$, passes through $(0,3)$: $y = x + 3$

BC: horizontal through $(-3,0)$: $y = 0$

12 The lines k and l are shown in the diagram. Find:

a The equation of line k, in general form.



$x - y - 3 = 0$

b The equation of line l, in general form.

$x + 2y - 9 = 0$

c The coordinates of point A.

$(0,4.5)$

d The coordinates of point C.

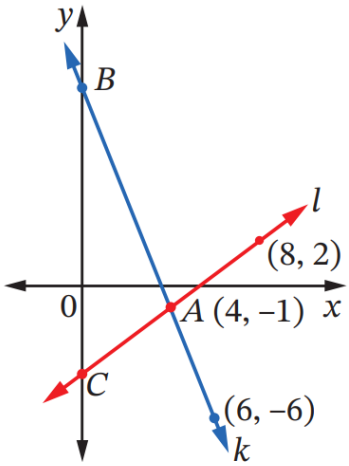
$(0,-3)$

e The area of ΔABC.

18.75 square units

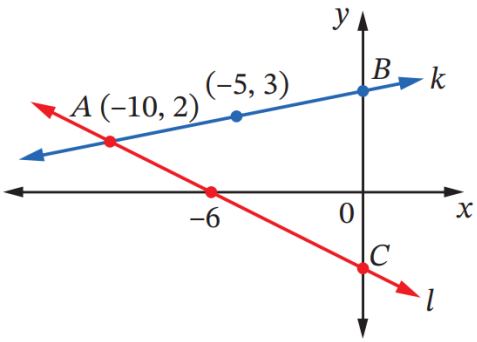
13 Find the area of ΔABC.

a



26 square units

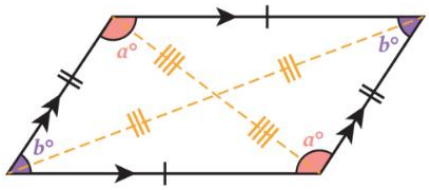
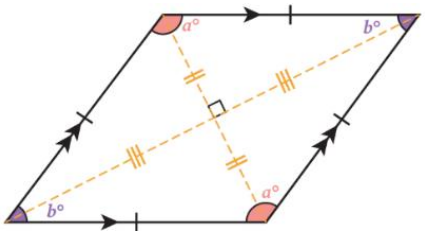
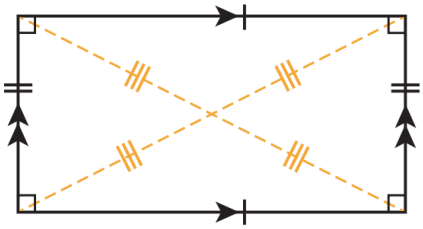
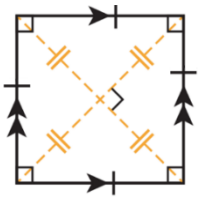
b



35 square units

Coordinate Geometry Problems: Quadrilaterals

ⓘ Types of Quadrilaterals and their Properties

Type of Quadrilateral	Definitions
Parallelogram 	Two pairs of opposite sides parallel **or** The opposite sides are equal, **or** One pair of opposite sides are equal and parallel, **or** The diagonals bisect each other
Rhombus 	All sides are equal **or** Two pairs of opposite sides parallel *and* one pair of adjacent sides equal, **or** The diagonals bisect each other at right angles
Rectangle 	Two pairs of opposite sides parallel *and* one angle a right angle **or** The diagonals are equal and bisect each other
Square 	Two pairs of opposite sides parallel **and** one pair of adjacent sides equal **and** one angle a right angle

ⓘ Diagonals Bisect Each Other

Two diagonals bisect (cut each other in half) when their midpoints are the same point.

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Example Solve coordinate geometry problems involving quadrilaterals

Show that the quadrilateral with vertices $A(1,2), B(5,4), C(7,0)$ and $D(3,-2)$ is a rhombus.

Easiest way is to show that all four sides are equal.

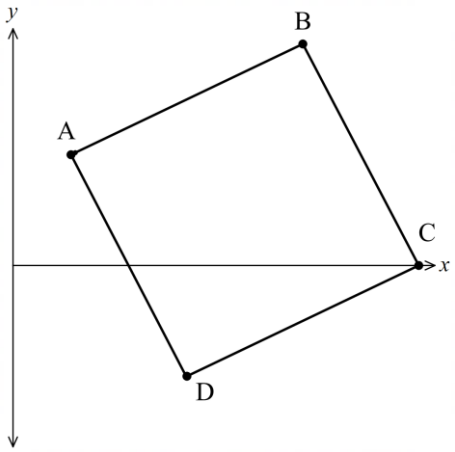
$AB = \sqrt{(5-1)^2 + (4-2)^2} = \sqrt{20}$

$BC = \sqrt{(7-5)^2 + (0-4)^2} = \sqrt{20}$

$CD = \sqrt{(3-7)^2 + (-2-0)^2} = \sqrt{20}$

$DA = \sqrt{(1-3)^2 + \big(2-(-2)\big)^2} = \sqrt{20}$

$\therefore$ ABCD is a rhombus (all sides equal)



Guided Practice

a Show that the quadrilateral with vertices E(1, 1), F(5, 3), G(7, −1) and H(3, −3) is a parallelogram by showing its diagonals bisect each other.

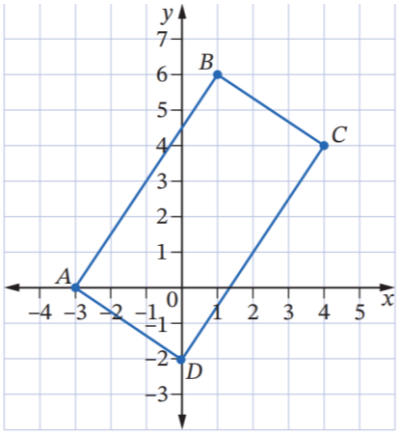
b Show that the quadrilateral with vertices P(1, 3), Q(5, 5), R(6, 3) and S(2, 1) is a rectangle by showing two pairs of opposite sides parallel and one pair of adjacent sides perpendicular.

1 $A(-3,0), B(1,6), C(4,4)$ and $D(0,-2)$ are the vertices of a rectangle.

a Find the lengths of AC and BD and show that the diagonals are equal.

b Find the midpoints of the diagonals AC and BD.

c What property of rectangles have you shown?



$$AC = \sqrt{(4-(-3))^2 + (4-0)^2} = \sqrt{65}$$

$$BD = \sqrt{(0-1)^2 + (-2-6)^2} = \sqrt{65}$$

Diagonals are equal.

$$\text{Midpoint of AC} = \left(\frac{-3+4}{2}, \frac{0+4}{2}\right) = \left(\frac{1}{2}, 2\right)$$

$$\text{Midpoint of BD} = \left(\frac{1+0}{2}, \frac{6+(-2)}{2}\right) = \left(\frac{1}{2}, 2\right)$$

The diagonals of a rectangle are equal and bisect each other.

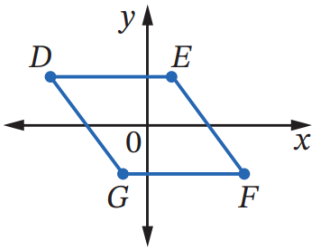
2 The vertices of a rhombus are $D(-4,2), E(1,2), F(4,-2)$ and $G(-1,-2)$

a Show that all side lengths are equal.

b By finding the gradients, show that the opposite sides are parallel.

c Show that the diagonals DF and GE are perpendicular.

d Show that the midpoints of DF and GE intersect at the same point (diagonals bisect).



$$DE = \sqrt{(1-(-4))^2 + (2-2)^2} = 5$$

$$EF = \sqrt{(4-1)^2 + (-2-2)^2} = 5$$

$$FG = \sqrt{(-1-4)^2 + (-2-(-2))^2} = 5$$

$$GD = \sqrt{(-4-(-1))^2 + (2-(-2))^2} = 5$$

All sides equal.

$$m_{DE} = \frac{2-2}{1-(-4)} = 0$$

$$m_{EF} = \frac{-2-2}{4-1} = -\frac{4}{3}$$

$$m_{FG} = \frac{-2-(-2)}{-1-4} = 0$$

$$m_{GD} = \frac{2-(-2)}{-4-(-1)} = -\frac{4}{3}$$

$m_{DE} = m_{FG}$ and $m_{EF} = m_{GD}$, so opposite sides are parallel.

$$m_{DF} = \frac{-2-2}{4-(-4)} = -\frac{1}{2}$$

$$m_{GE} = \frac{2-(-2)}{1-(-1)} = 2$$

$$m_{DF} \times m_{GE} = -\frac{1}{2} \times 2 = -1$$

Diagonals are perpendicular.

$$\text{Midpoint of DF} = \left(\frac{-4+4}{2}, \frac{2+(-2)}{2}\right) = (0,0)$$

$$\text{Midpoint of GE} = \left(\frac{-1+1}{2}, \frac{-2+2}{2}\right) = (0,0)$$

Diagonals bisect each other.

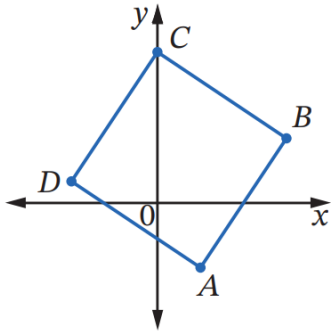
3 A square has vertices $A(2,-3), B(6,3), C(0,7)$ and $D(-4,1)$

a Show that the diagonals are equal.

b Show that its diagonals meet at right angles.

c Show that the diagonals bisect each other.

d Hence explain why $ABCD$ must be a square.



$$AC = \sqrt{(0-2)^2 + (7-(-3))^2} = \sqrt{104} = 2\sqrt{26}$$

$$BD = \sqrt{(-4-6)^2 + (1-3)^2} = \sqrt{104} = 2\sqrt{26}$$

Diagonals are equal.

$$m_{AC} = \frac{7-(-3)}{0-2} = -5$$

$$m_{BD} = \frac{1-3}{-4-6} = \frac{1}{5}$$

$$m_{AC} \times m_{BD} = -5 \times \frac{1}{5} = -1$$

Diagonals meet at right angles.

$$\text{Midpoint of AC} = \left(\frac{2+0}{2}, \frac{-3+7}{2}\right) = (1,2)$$

$$\text{Midpoint of BD} = \left(\frac{6+(-4)}{2}, \frac{3+1}{2}\right) = (1,2)$$

Diagonals bisect each other.

ABCD has equal diagonals that bisect each other at right angles, so ABCD is a square.

4 Consider a quadrilateral $ABCD$. $A(4,9), B(0,4), C(-2,3)$ and $D(2,8)$.

a Find the midpoint of the interval joining $A(4,9)$ and $C(-2,3)$.

b Find the midpoint of the interval joining $B(0,4)$ and $D(2,8)$.

c What can you conclude about the diagonals of the quadrilateral?

d Explain why $ABCD$ must be a parallelogram.

$$\text{Midpoint of AC} = \left(\frac{4+(-2)}{2}, \frac{9+3}{2}\right) = (1,6)$$

$$\text{Midpoint of BD} = \left(\frac{0+2}{2}, \frac{4+8}{2}\right) = (1,6)$$

The diagonals bisect each other.

Since the diagonals bisect each other, ABCD is a parallelogram.

5 The points $A(3,1)$, $B(10,2)$, $C(5,7)$ and $D(-2,6)$ are the vertices of a quadrilateral.

a Find the lengths of all four sides.

b Explain why ABCD is a rhombus.

$$AB = \sqrt{(10-3)^2 + (2-1)^2} = \sqrt{50}$$

$$BC = \sqrt{(5-10)^2 + (7-2)^2} = \sqrt{50}$$

$$CD = \sqrt{(-2-5)^2 + (6-7)^2} = \sqrt{50}$$

$$DA = \sqrt{(3-(-2))^2 + (1-6)^2} = \sqrt{50}$$

All four sides are equal, so ABCD is a rhombus.

6 Show that $A(2,5)$, $B(3,7)$, $C(-4,-1)$ and $D(-5,2)$ do not form a parallelogram.

$$\text{Midpoint of AC} = \left(\frac{2+(-4)}{2}, \frac{5+(-1)}{2}\right) = (-1,2)$$

$$\text{Midpoint of BD} = \left(\frac{3+(-5)}{2}, \frac{7+2}{2}\right) = \left(-1, \frac{9}{2}\right)$$

Diagonals do not bisect each other, so ABCD is not a parallelogram.

7 The quadrilateral ABCD has vertices $A(-1,1), B(3,-1), C(5,3)$ and $D(1,5)$.

a Show that the opposite sides are parallel, and hence that ABCD is a parallelogram.

b Show that AB $\perp$ BC, and hence that ABCD is a rectangle.

c Show that AB = BC, and hence that ABCD is a square.

$$m_{AB} = \frac{-1-1}{3-(-1)} = -\frac{1}{2}$$

$$m_{DC} = \frac{5-3}{1-5} = -\frac{1}{2}$$

$$m_{BC} = \frac{3-(-1)}{5-3} = 2$$

$$m_{AD} = \frac{5-1}{1-(-1)} = 2$$

$m_{AB} = m_{DC}$ and $m_{BC} = m_{AD}$, so opposite sides are parallel.

ABCD is a parallelogram.

$$m_{AB} \times m_{BC} = -\frac{1}{2} \times 2 = -1$$

AB $\perp$ BC, so ABCD is a rectangle (parallelogram with adjacent sides perpendicular)

$$AB = \sqrt{(3-(-1))^2 + (-1-1)^2} = \sqrt{20}$$

$$BC = \sqrt{(5-3)^2 + (3-(-1))^2} = \sqrt{20}$$

AB = BC, so ABCD is a square (rectangle with adjacent sides equal)

8 A quadrilateral, $ABCD$, has vertices $A(2,7), B(4,9), C(8,5)$ and $D(6,3)$.

a Find the gradient of:

a AB	**b** BC	**c** CD	**d** DA
1	-1	1	-1

b What type of quadrilateral is $ABCD$?

rectangle

c Explain why.

Opposites are parallel and adjacent sides are perpendicular

9 Find the gradients of these four lines and explain why the quadrilateral formed by the four lines must be a rectangle.

$4x - 3y + 10 = 0, \quad 3x + 4y + 7 = 0, \quad 4x - 3y - 7 = 0, \quad 3x + 4y + 1 = 0$

$\text{Line 1: } 4x - 3y + 10 = 0 \rightarrow m_1 = \frac{4}{3}$

$\text{Line 2: } 3x + 4y + 7 = 0 \rightarrow m_2 = -\frac{3}{4}$

$\text{Line 3: } 4x - 3y - 7 = 0 \rightarrow m_3 = \frac{4}{3}$

$\text{Line 4: } 3x + 4y + 1 = 0 \rightarrow m_4 = -\frac{3}{4}$

$m_1 = m_3$ and $m_2 = m_4$, so opposite sides are parallel.

$m_1 \times m_2 = \frac{4}{3} \times \left(-\frac{3}{4}\right) = -1$, so adjacent sides are perpendicular.

The quadrilateral is a rectangle.

10 A quadrilateral has vertices $P(-7,2), Q(2,-7), R(5,-4)$ and $S(-4,5)$.

a Find the lengths of PR and QS in surd form.

b Find the midpoints of PR and QS.

c Is PR perpendicular to QS? Explain why.

d What type of quadrilateral is $PQRS$? Explain.

$PR = \sqrt{(5-(-7))^2 + (-4-2)^2} = \sqrt{180}$

$QS = \sqrt{(-4-2)^2 + (5-(-7))^2} = \sqrt{180}$

$\text{Midpoint of PR} = \left(\frac{-7+5}{2}, \frac{2+(-4)}{2}\right) = (-1,-1)$

$\text{Midpoint of QS} = \left(\frac{2+(-4)}{2}, \frac{-7+5}{2}\right) = (-1,-1)$

$m_{PR} = \frac{-4-2}{5-(-7)} = -\frac{1}{2}$

$m_{QS} = \frac{5-(-7)}{-4-2} = -2$

PR is not perpendicular to QS (product of gradients ≠ -1).

PQRS is a rectangle (not a rhombus)

The diagonals PR and QS are equal length and bisect each other at $(-1,-1)$

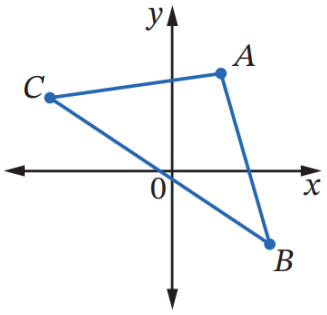
11 A quadrilateral has vertices $C(2,6), D(-5,2), E(-1,-5)$ and $F(6,-1)$

a Find the length of each diagonal.

b Find the midpoint of each diagonal.

c Show that the diagonals are perpendicular.

d What type of quadrilateral is $PQRS$? Explain.



$$CE = \sqrt{(-1-2)^2 + (-5-6)^2} = \sqrt{130}$$

$$DF = \sqrt{(6-(-5))^2 + (-1-2)^2} = \sqrt{130}$$

$$\text{Midpoint of } CE = \left(\frac{2+(-1)}{2}, \frac{6+(-5)}{2}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$\text{Midpoint of } DF = \left(\frac{-5+6}{2}, \frac{2+(-1)}{2}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$m_{CE} = \frac{-5-6}{-1-2} = \frac{11}{3}$$

$$m_{DF} = \frac{-1-2}{6-(-5)} = -\frac{3}{11}$$

$$m_{CE} \times m_{DF} = \frac{11}{3} \times \left(-\frac{3}{11}\right) = -1$$

Therefore diagonals are perpendicular.

CDEF is a rhombus (diagonals equal length, bisect each other, and are perpendicular)

12 $A(2,4), B(4,-3)$, and $C(-5,3)$ are the vertices of a triangle.

a X and Y are midpoints of AB and AC respectively.
Find the coordinates of X and Y.

b Find the gradients of XY and CB.

c What type of quadrilateral is $XYBC$?

d Find the lengths of XY and CB. Hence show that $CB = 2XY$.

$$\text{X} = \left(\frac{2+4}{2}, \frac{4+(-3)}{2}\right) = \left(3, \frac{1}{2}\right)$$

$$\text{Y} = \left(\frac{2+(-5)}{2}, \frac{4+3}{2}\right) = \left(-\frac{3}{2}, \frac{7}{2}\right)$$

$$\text{Gradient XY} = \frac{\frac{7}{2}-\frac{1}{2}}{-\frac{3}{2}-3} = -\frac{2}{3}$$

$$\text{Gradient CB} = \frac{3-(-3)}{-5-4} = -\frac{2}{3}$$

XY and CB are parallel, so XYBC is a trapezium

$$\text{Length XY} = \sqrt{\left(-\frac{3}{2}-3\right)^2 + \left(\frac{7}{2}-\frac{1}{2}\right)^2} = \frac{3\sqrt{13}}{2}$$

$$\text{Length CB} = \sqrt{(-5-4)^2 + (3-(-3))^2} = 3\sqrt{13}$$

$$\therefore CB = 2XY$$

13 The quadrilateral $ABCD$ has vertices $A(1,-4), B(3,2), C(-5,6)$ and $D(-1,-2)$.

a Find the midpoints P of AB, Q of BC, R of CD, and S of DA.

b Prove that PQRS is a parallelogram by showing that PQ ∥ RS and PS ∥ QR.

$$P = \left(\frac{1+3}{2}, \frac{-4+2}{2}\right) = (2,-1)$$

$$Q = \left(\frac{3+(-5)}{2}, \frac{2+6}{2}\right) = (-1,4)$$

$$R = \left(\frac{-5+(-1)}{2}, \frac{6+(-2)}{2}\right) = (-3,2)$$

$$S = \left(\frac{-1+1}{2}, \frac{-2+(-4)}{2}\right) = (0,-3)$$

$$m_{PQ} = \frac{4-(-1)}{-1-2} = -\frac{5}{3}$$

$$m_{RS} = \frac{-3-2}{0-(-3)} = -\frac{5}{3}$$

$PQ \parallel RS$ since $m_{PQ} = m_{RS}$

$$m_{PS} = \frac{-3-(-1)}{0-2} = -\frac{2}{-2} = 1$$

$$m_{QR} = \frac{2-4}{-3-(-1)} = 1$$

$PS \parallel QR$ since $m_{PS} = m_{QR}$

PQRS is a parallelogram since opposite sides are parallel.

14 Explain why the four lines $l_1: y = x+1, l_2: y = x-3, l_3: y = 3x+5$ and $l_4: y = 3x-5$ enclose a parallelogram. Then find the vertices of this parallelogram.

$l_1 \parallel l_2$ and $l_3 \parallel l_4$, so opposite sides are parallel

The four lines enclose a parallelogram

Solve simultaneously to find point of intersection:

$l_1 = l_3: x+1 = 3x+5$

$x = -2, y = -1$

Vertex $(-2,-1)$

$l_1 = l_4: x+1 = 3x-5$

$x = 3, y = 4$

Vertex $(3,4)$

$l_2 = l_3: x-3 = 3x+5$

$x = -4, y = -7$

Vertex $(-4,-7)$

$l_2 = l_4: x-3 = 3x-5$

$x = 1, y = -2$

Vertex $(1,-2)$

Vertices: $(-2,-1), (3,4), (-4,-7), (1,-2)$